How Much PENCIL Can You Afford?

Compute-allocation laws for link prediction · choosing shortlist k and context budget s without retraining

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WHAT WE ARE DOING

We rank every candidate link with Adamic–Adar, run PENCIL only on the top-k candidates, and vary PENCIL's sampled subgraph budget s. **The (k, s) sweep measures how much accuracy remains at each GPU budget. The model is not retrained.**

1 AA rank — CPU <1 ms → 2 top-k — shortlist 0–100% → 3 PENCIL — re-rank with s

FREE-GPU REPRODUCTION · PAPER USED 4–8× A100

44.00	50.46	112
Cora test	Citeseer test	score dumps
paper 42.23	paper 47.51	4 regimes

EXPERIMENT SETUP

Data	Cora and Citeseer
Tests	shared negative pool and HeaRT hard negatives
Selection	choose (k, s) on validation under fixed compute caps
Checks	corrected score-precision and shortlist tie bias

WHAT WE MEASURE

Test MRR ×100 · measured T4 GPU time · peak VRAM · three random seeds per budget

RESULTS FROM THE BUDGET SWEEP

01 Half fanout is enough in 3 of 4 regimes

At s=2×10 (~65% GPU), PENCIL matches Cora Pool (42.9 vs 43.7) and beats full s=2×20 on Citeseer Pool (51.4 vs 49.7) and Cora HeaRT (14.3 vs 13.3). Citeseer HeaRT is the exception: 14.2 vs 17.8.

02 A ≤10% GPU hybrid beats heuristics everywhere

Validation-selected (k, s) improves over Adamic–Adar in all four tests: 33.1→38.4 on Cora Pool, 31.5→34.7 on Citeseer Pool, 12.0→12.8 on Cora HeaRT, and 11.0→13.8 on Citeseer HeaRT.

03 PENCIL pays off where graph structure is blind

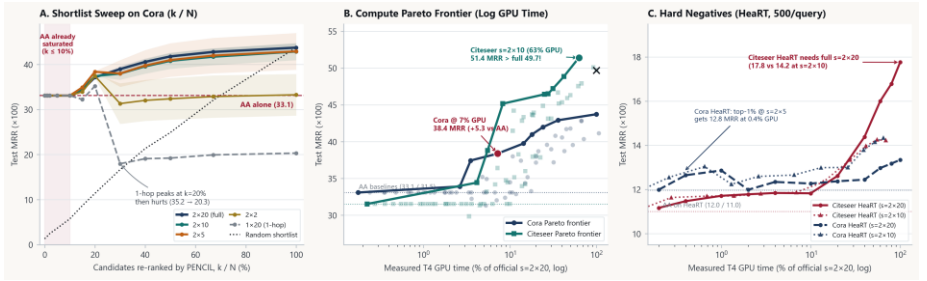
Re-ranking only the top 10% changes almost nothing because AA already ranks easy links highly. The useful compute lies in the zero-common-neighbour tail: 57.3% of true Cora test links have zero shared neighbours.

04 Cross-graph rules can beat noisy local selection

Applying Cora's validation-chosen rule unchanged to Citeseer beats Citeseer's own validation pick at every cap ≤50%; at the 25% cap, transferred MRR is 44.9 versus 37.9 from local selection.

Main result: reducing fanout is usually safer than shortening the candidate list.

BUDGET SWEEPS



Shortlist saturation · compute Pareto frontiers · HeaRT hard-negative stress test

VALIDATION-SELECTED OPERATING POINTS · TEST MRR ×100

Policy	Cora Pool	Citeseer Pool	Cora HeaRT	Citeseer HeaRT
Chance	1.3	1.5	1.4	1.4
AA · 0% GPU	33.1	31.5	12.0	11.0
Hybrid · ≤10%	38.4 · 7%	34.7 · 9%	12.8 · 0.4%	13.8 · 8%
Half · s=2×10	42.9 · 67%	51.4 · 63%	14.3 · 65%	14.2 · 69%
Full · s=2×20	43.7	49.7	13.3	17.8

ADDITIONAL RESULTS

Cora → Citeseer At the 25% cap, the transferred rule reaches 44.9 MRR, beating Citeseer's own noisy validation pick (37.9).

Peak VRAM · Cora HeaRT 2,219 MB (2×20) → 1,347 MB (2×10) → 558 MB (2×2).

All operating points are selected on validation data; test data are used only for the reported results.